SIMATS ENGINEERING

ASSESSMENT TOOL 3 – CASE STUDY

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Course Name	INTERNET PROGRAMMING
Course Code	CSA43
Course Outcome Covered	CO-1 (L2): Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, a
Assessment Tool	Case Study
Weightage	10%
Total Marks	10

CASE STUDY: Web Development Standards Implementation

A start-up is developing a modern web application. During development, the team encounters issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols. The observations include embedding a promotional video, securely sending registration data, using semantic navigation, correcting XML configuration syntax, and selecting the correct web communication protocol.

Question 1: Select the correct HTML tag for video embedding and justify your answer.

Answer: B. <video src="video.mp4" controls></video>

The HTML5 **<video>** element is the standard semantic element for embedding video content in a web page. The **src** attribute specifies the video file and **controls** provides browser controls such as play, pause, volume and seeking. The other options use non-standard elements for HTML5 video embedding.

Question 2: Identify the appropriate HTTP method for form submission and explain why.

Answer: B. POST

POST is appropriate when a registration form sends user data to the server because the submitted values are placed in the HTTP request body instead of the URL. Registration normally creates or changes server-side data, so POST is more suitable than GET. For sensitive information, the form should also be transmitted over HTTPS.

Example:

```
<form action="/register" method="POST">
  <input type="text" name="name" required>
  <button type="submit">Register</button>
</form>
```

Question 3: Choose the proper semantic element for navigation and justify its usage.

Answer: B. <nav>

The **<nav>** element is the semantic HTML element intended for a section containing navigation links. It gives the document a meaningful structure and helps browsers, search engines and assistive technologies understand that the links form a navigation area. A generic ** ** does not provide this semantic meaning.

Example:

```
<nav aria-label="Main navigation">
  <a href="/">Home</a>
  <a href="/events">Events</a>
  <a href="/contact">Contact</a>
</nav>
```

Question 4: Identify the correct XML syntax and explain the error in other options.

Answer: B. <note><to>User</to><from>Admin</from></note>

Option B is correct because XML elements are properly opened and closed, and the child elements are correctly nested inside the root **<note>** element.

- **Option A:** The root element **<note>** is not closed with **</note>**.
- **Option C:** The **<to>** element is opened but incorrectly closed using **</from>**; XML requires matching start and end tags.
- **Option D:** The **<note>** start tag is incomplete because the closing **>** is missing.

Question 5: Choose the correct protocol for web communication and justify your choice.

Answer: B. HTTP (HyperText Transfer Protocol)

HTTP is the application-layer protocol used for communication between web browsers and web servers. It supports operations such as requesting HTML pages, submitting forms, and transferring web resources. HTTPS is the secure version of HTTP and should be used when data such as registration information must be protected in transit.

Final Answers at a Glance

Question	Correct Answer	Key Reason
1	B – <video>	Standard HTML5 element for video embedding
2	B – POST	Sends form data in the request body; suitable for registration
3	B – <nav>	Semantic element for navigation links
4	B – Correct XML	All elements are correctly opened, nested and closed
5	B – HTTP	Protocol used for browser–web server communication

Conclusion

The case study demonstrates the correct application of HTML5 semantic elements, HTTP methods, XML well-formedness rules, and web communication protocols. Following these standards improves correctness, accessibility, maintainability and reliable communication between the browser and server.

Code of Conduct

I, **BILLE VINAY KUMAR (Reg. No: 192373042)**, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____